N-Glycans as fundamental immune-checkpoints at the frontiers of inflammation, autoimmunity and cancer

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DOI: 10.1039/9781837676101-00001

Glycans cover the surface of essentially all cells, giving rise to the so-called glycocalyx. Glycans, and particularly *N*-glycans are key players in defining the fate and the nature of the immune response, participating in both pro-inflammatory and anti-inflammatory pathways occurring in inflammation, autoimmunity and cancer. Glycans and glycan-binding proteins (GBPs) are instrumental molecules that integrate the immunological landscape in homeostasis, being involved in how the host immune system perceives the self and non-self. Perturbations in these immune regulatory circuits created by glycans and GBPs may result in the loss of immune tolerance associated with autoimmunity and inflammatory conditions, but also in malignant transformation. This chapter will discuss the regulatory role played by *N*-glycans as fundamental immune-checkpoints in the interplay between the immune cells and their surrounding microenvironment associated with homeostasis and disease. Furthermore, we will also explore how *N*-glycans can act as promising diagnostic and prognostic biomarkers in disease, as well as appealing targets for new therapeutic approaches in inflammation, autoimmunity and cancer.

1 Introduction

Glycans, also recognized as monosaccharides or polysaccharides, are carbohydrate structures that can be found intracellularly and extracellularly at the cell surface of essentially all cells.[1,2] Protein glycosylation occurs in the Golgi and endoplasmic reticulum and comprises two major classes of glycoproteins according to the glycan-protein linkage: *N*-glycans (sugar molecules attached to an asparagine residue of proteins *via* the nitrogen atom) and *O*-glycans (sugar moieties attached *via* the oxygen atom to a serine or threonine residue of a nascent protein). This posttranslational modification is fundamental for numerous cellular and molecular biological processes, such as cell interaction and adhesion, cell-fate decisions, cell migration, regulation of receptor activation and

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signal transduction, embryonic development, distinction between "self" and "non-self" and regulation of the immune system.[3–5] Furthermore, glycans can act as ligands for glycan-binding proteins (GBPs), including galectins (Gal), C-type lectins (CTLs), and selectins, highlighting their role as key components in immune-regulatory circuits and in the balance between immune activation and immune suppression. Therefore, changes in glycan expression have remarkable implications in the homeostatic environment associated with the pathogenesis of major diseases.[6] These changes can be triggered by different factors, including altered glycan biosynthesis, changes in the expression of substrate proteins and altered activity of enzymes essential in the glycosylation pathway.[7] Several studies reported that alterations in glycosylation are associated with immune dysregulation in different pathological conditions, including inflammatory bowel disease (IBD),[8–11] autoimmune disorders,[12–14] infection[15–17] and cancer.[3,18–21] In this chapter, we will focus on the *N*-glycosylation pathway, in both immune and epithelial cells, addressing how *N*-glycans regulate immune response in health and in disease.

2 Glycans as immune regulatory circuits in homeostasis

Glycobiology has a long history initiated with the discovery of the nature of carbohydrate structures through research in the human ABO (H) and Lewis blood groups.[22,23] Over the last decades, glycobiology has gained increased importance in the immunology field, with several evidence demonstrating the pivotal role of glycans and GBPs in regulating both molecular and cellular mechanisms that govern the immune homeostasis, including innate and adaptive immune responses.[5,24–27]

Glycans play crucial roles in homeostasis of both T and B cells, even from the early stages of their development, as reviewed by us.[28] Highlighting some of the findings regarding *N*-glycosylation, complex *N*-glycans are paramount for T cell receptor (TCR) selection, by inhibiting death by neglect and by enhancing TCR sensitivity to low-affinity peptide major histocompatibility complex (MHC).[29] Recently, our group disclosed stage-specific glycome compositions during T cell development of both human and mouse thymocytes. The expression of complex *N*-glycans was shown to impact ß-selection and TCR repertoire diversity. Moreover, deletion of complex *N*-glycans favored γδ T cell development and impaired regulatory T cell (Tregs) expansion.[30]

Besides T cell development, *N*-glycans also modulate T cell activity and function.[31] Barbosa and colleagues demonstrated that blockage of *N*-linked glycans on human leukocyte antigen (HLA) class I (homolog of MHC-I in mice) led to protein intracellular denaturation, resulting in decreased levels of the protein on cell surface.[32] Moreover, antigen-presenting cells (APCs) lacking complex *N*-glycans were shown to exhibit decreased MHC-II binding and impaired antigen presentation to T cells.[33] Glycans are also important in TCR signaling. In fact, Demetriou and colleagues showed that β1,6- *N*-Acetylglucosamine (GlcNAc)–branched

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N-glycans catalyzed by *N*-acetylglucosaminyltransferase-V (GnT-V) (encoded by the *MGAT5* gene) regulate the TCR signaling and activation. These carbohydrates can be further elongated to form poly-*N*-acetyllactosamine (poly-LacNAc) structures, acting as ligands for Gal-3, creating a molecular lattice, that modulates TCR signaling and impacts the threshold for T cell activation.[34] Similarly, mice lacking the β1,3-*N*-acetylglucosaminyltransferase (B3GnT2), resulting in less poly-LacNAc structures on *N*-glycans, displayed an increased TCR clustering, as well as an hyperactivation and proliferation of both T and B cells.[35] Glycans represent an important layer of information for the recognition by environmental players, namely by GBPs, which are crucial players in the immune-mediated response.[36] Particularly in the context of T cell activity and function, Gal-1, Gal-3 and Gal-9 present immunosuppressive effects on T cells. In particular, Gal-3 presents a dual effect on T cells, depending if its expression is intra- or extracellular. The former leads to blockage of apoptosis, while the latter, through binding to glycosylated receptors on the T cell surface, leads to inhibition of activated T cells.[37–40] Gal-1 and Gal-3 bind preferentially to poly-LacNAc motifs and this assembly leads to the inhibition of transendothelial migration and induces apoptosis of T cells.[41]

Glycosylation of the surface protein CD28 can also regulate T cell adhesion and stimulation. Mutations in the *N*-glycosylation sites of CD28 resulted in increased binding to CD80.[42] Additionally, a reduction in branched *N*-glycans led to a decrease in the surface expression and retention of CD25 and hampered IL-2 binding, ultimately preventing Treg activation and promoting T helper (Th) 17 differentiation.[43] In fact, cytokines regulate T cell *N*-glycome composition. In resting T cells, IL-2 signaling decreased branching by upregulating *Mgat1* and downregulating *Mgat5*. Since *Mgat1*-encoded enzyme (GnT-I) has higher affinity to UDP-GlcNAc, the limitation of this substrate resulted in reduced branching. Conversely, in activated T cells, TCR signaling elevated *Mgat5* expression resulting in the promotion of branching *N*-glycosylation.[44,45] The opposite effects in resting *versus* activated T cells suggest a mechanism of regulating activation and self-tolerance mediated by complex *N*-glycans. Opposite effects on T cell dynamics are described for other glycosylation alterations. For instance, core fucosylation, which is catalyzed by α1,6-fucosyltransferase (Fut8) enzyme (encoded by *FUT8* gene), was also shown to impact T cell activation.[46,47] On the other hand, sialylated antigens induced immune-suppression by inhibiting effector T cell proliferation.[48,49]

B cell development is also highly regulated by *N*-glycans,[28] despite remaining so far underexplored. As example, core fucosylation impacts pre-B cell development by interfering with pre-B cell receptor (BCR) assembly affecting B cell intracellular signaling and proliferation.[50] Additionally, B cell-specific deletion of *Mgat1* gene resulted in decreased surface expression of the BCR, hampering positive selection. Consistently, the deficiency in complex *N*-glycans on B cells was reported to be essential for the survival of T2 transitional peripheral B cells, while not affecting mature B cells. These evidences suggest complex *N*-glycans as

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fundamental players in maintaining the homeostatic threshold of pre-BCR/BCR signaling, allowing positive selection.[51] Moreover, extrinsic sialylation by α2,6-sialyltransferase 1 (ST6GAL1) enzyme was found to be a significant pro-survival factor during B cell development, whereas B cell intrinsic ST6GAL1 was shown to be required for marginal zone (MZ) B cell development.[52]

B cell activity and function can also be modulated by *N*-glycans. Core fucosylation was described to affect the humoral response, with *Fut8*$^{-/-}$ mice showing reduced levels of IgG subclasses (but not IgM) due to low cytokine levels, such as interleukin (IL)-4, IL-5, IL-6, interferon (IFN)γ, and transforming growth factor (TGF) secreted by Th cells.[53] Upon B cell activation and differentiation, changes in the *N*-glycosylation patterns occur. Naive and memory B cells were reported to express GnT-V-mediated branched *N*-glycans at the surface, which contain poly-LacNAc motifs, serving as high-affinity binding sites for many galectins.[54,55] On the other hand, germinal center (GC) B cells were described to be modified with β1,6-GlcNAc- and galactose-containing disaccharides, precluding Gal-9 binding. Interestingly, Gal-9 was described to regulate BCR signaling, antagonizing naive B cell activation.[54,55] Additionally, the BCR can form a complex with immunoregulatory glycoproteins, including the sialic acid–binding immunoglobulin-type lectins (Siglec-2 or CD22), an inhibitory receptor mainly found on B cells, which modulate the BCR activation threshold. Notably, *ST6Gal1* deficient B cells showed suppressed BCR signaling, whereas mice deficient in both CD22 and its ligand (*Cd22*$^{-/-}$*St6gal1*$^{-/-}$) presented restored BCR signaling, demonstrating the importance of α2,6 sialic acids/Siglec axis in immunomodulation of BCR signaling.[56]

Furthermore, *N*-glycosylation is crucial for the effector functions of immunoglobulins (Ig). IgG, a glycoprotein secreted by antibody-secreting B cells, consists of a crystallizable fragment (Fc), responsible for triggering effector functions, and antigen-binding fragment (Fab) arms, responsible for antigen binding. The Fc region of IgG is covalently attached to a single bi-antennary *N*-glycan which plays a key structural role in stabilizing the conformation of the Fc region.[57] The Fc-linked *N*-glycan can comprise different glycoforms, generally combining the presence or absence of fucose, galactose, sialic acid and bisecting GlcNAc.[58] Although heterogeneous, the profile of IgG glycosylation is relatively constant, displaying high proportion of core fucosylation (96%), low bisecting (8%), sialylation (4%), and intermediate galactosylation (40%).[59,60] Changes in the composition of these glycans significantly affect the effector function of IgG, as observed by the loss of function in non-glycosylated IgGs.[61] Interestingly, galactosylation and sialylation levels decreased with age and displayed a significant gender dependence.[59] Moreover, cytokines in the microenvironment can regulate IgG glycan patterns by regulating intracellular glycosyltransferases in human B cells. Increased levels of IgG galactose were observed in the presence of IFNγ, whereas increased levels of IgG sialylation were observed upon IL-21 and IL-17A stimulation.[62]

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3 Sweet mediators of inflammation: changes in *N*-glycans as potential triggers of intestinal inflammation and disease progression

Increasing evidence indicates that changes in gut glycosylation may occur before the onset of disease, during the transition from health to inflammatory conditions.[63–65] However, the mechanisms governing these glycosylation changes are still poorly explored. Intestinal inflammation can be triggered by the disruption of the intestinal barrier and/or the mucosal immunity, leading to chronic inflammatory conditions. IBD is a chronic inflammatory disorder of the gastrointestinal tract that encompasses Crohn's disease (CD) and ulcerative colitis (UC).[66] The causes contributing to the development of IBD are complex, being associated with a dysregulated immunologic response, alterations in the intestinal microbiome, genetic and environmental factors. Recently, the aberrant expression of glycans has emerged as a novel mechanism in the pathogenicity of IBD.[67] Herein, we will describe the most common *N*-glycosylation changes contributing to inflammation onset and disease progression, focusing on IBD (see Fig. 1).

3.1 Sweet molecules, bitter consequences: intestinal *N*-glycans as shapers of chronic inflammation

Intestinal mucosa is the major physical and biological barrier that integrates different stimuli, supporting host-microbe interactions and immune responses. Intestinal epithelial cells (IECs) express a diverse repertoire of *O*- and *N*-glycans that maintain mucosal homeostasis by shaping intestinal inflammation in the host.[68] Interestingly, in homeostatic conditions, the bottom of the intestinal crypts in the small intestine (SI) is characterized by an enriched stem cell zone abundant in mannose-enriched glycoproteins.[69] As we move towards the top of the crypts, not only cells become more differentiated, but also the complexity of *N*-glycans expressed by these cells increases, with a predominance of complex-type *N*-glycans bearing terminal LacNAc.[70] This LacNAc expression was shown to be crucial for the control of stem cell behavior and proliferation,[69] possibly affecting the epithelial recovery after chronic inflammation of the tissue.

In chronic intestinal inflammation, the most well-described glycosylation alteration is the increased expression of immature and truncated glycans.[71] These alterations promote inflammation onset that perpetuates upon failure of tissue repair, contributing to disease pathogenesis. Additionally, in the intestinal epithelium, FUT1 and FUT2 mediate the expression of the major fucosylation class in humans and mice – the α1,2-linked fucose.[68,72,73] Among the genetic and epigenetic factors involved in intestinal inflammatory dysregulation, *FUT1* and *FUT2* polymorphisms have been associated with CD susceptibility.[74,75] Specifically, *FUT2,* which encodes for α1,2-fucosyltransferase, is responsible for the secretion of ABO blood group antigens into mucosal surfaces. One of the most studied variants, *FUT2 W143X*, introduces a nonsense mutation

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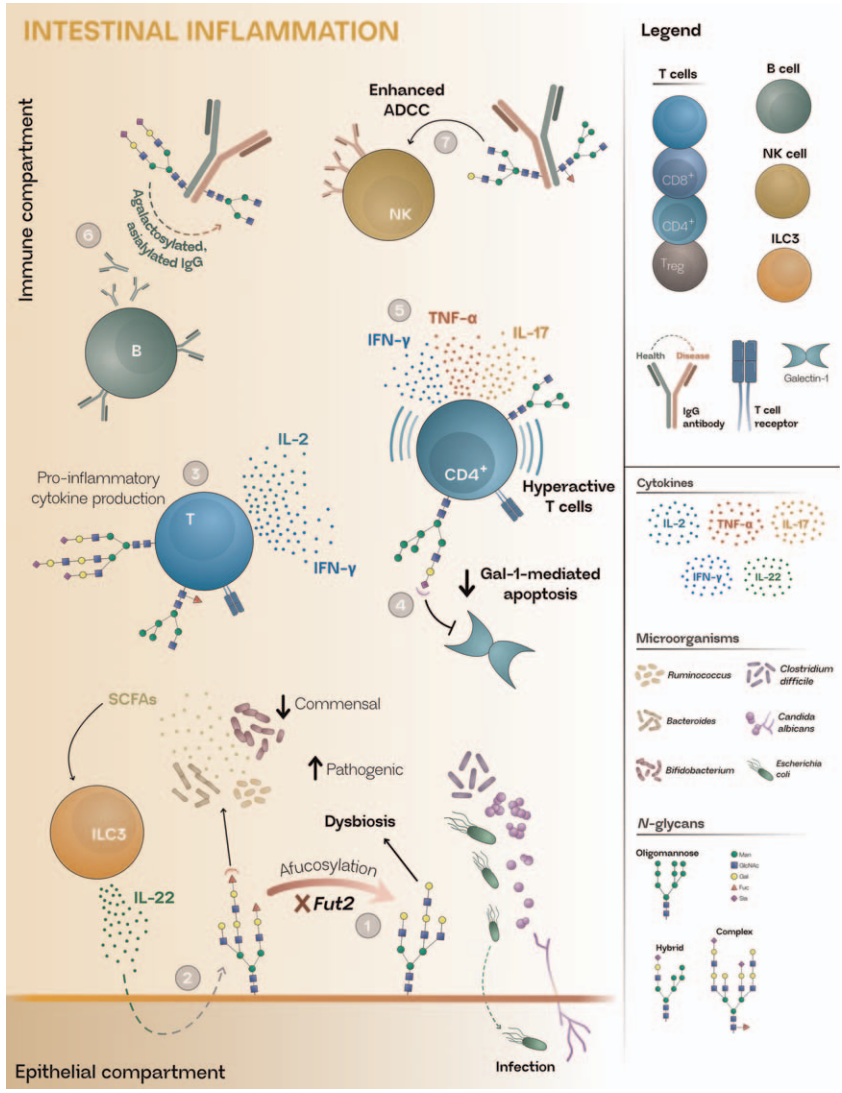


Fig. 1 Changes in mucosa glycosylation perturb the symbiotic relationship between host and microbiome culminating in dysbiosis associated with intestinal inflammation. Cellular *N*-glycosylation undergoes dynamic alterations during the transition from health to pathological conditions, being crucial molecules involved in the regulation of pro-inflammatory responses associated with inflammation. (1) Alterations in Fut2 expression, promoting the expression of afucosylated forms, have been associated with dysbiosis. (2) Production of IL-22 by ILCs induce Fut2 upregulation on intestinal epithelial cells (IECs). (3) Mucosal T cells from inflamed tissue of IBD patients revealed increased levels of core fucosylation associated with the production of inflammatory cytokines. (4) UC patients display elevated levels of α2,6-sialylation in colonic tissue. Upregulation of ST6GAL1, detected in colonic biopsies of IBD patients, masks T cells from Gal-1, inhibiting its binding and thereby inducing inflammation. (5) UC patients present a deficiency in β1,6-GlcNAc–branched *N*-glycans on colonic $CD4^+$ T cells, associated with T cells hyperactivation and production of pro-inflammatory cytokines. (6) Inflamed IgG display an agalactosylated and asialylated profile which promotes the pro-inflammatory environment. Agalactosylated IgG was shown to enhance antibody-dependent phagocytosis. CD patients present a decrease in IgG galactose associated either with the downregulation of the β4GalT gene in B cells or SNPs in *MGAT5* gene observed in UC patients. (7) Core fucosylation and bisecting of the Fc region enhance antibody-dependent cell-mediated cytotoxicity (ADCC).

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that results in a nonfunctional enzyme, leading to a non-secretor phenotype. This ABO non-secretory phenotype, whose prevalence is higher in Asian populations (50%) compared to European populations (approximately 20%), has been linked to altered gut microbiota composition and increased susceptibility to gastrointestinal infections and inflammatory diseases, including CD.[74]

Moreover, *FUT2* mutations were described to also contribute to colitis-associated microbial dysbiosis in CD patients.[76] Moreover, in IBD patients, the expression of FUT2 and α1,2-linked fucose were decreased.[77] Additionally, *Fut2*-deficient mice showed increased susceptibility to pathogenic inflammation and severe colitis upon dextran sodium sulfate (DSS) treatment.[78] After fucose metabolic supplementation of these mice, an attenuated inflammation was observed upon DSS-induced colitis.[79] Of note, *FUT2* polymorphisms are also associated with other acute and chronic inflammatory disorders, including type I diabetes and psoriasis.[80,81] Together, these evidences strongly corroborate the seminal role of epithelial fucosylation in controlling intestinal homeostasis by regulating host defense against pathogenic infection.

Despite being highly *O*-glycosylated, mucins are also decorated with *N*-glycans, important for the function of these glycoproteins. Interestingly, mass spectrometry-based *N*-glycan of mouse colonic Muc2 demonstrated that the binding of dendritic cells (DC)-associated C-type lectin (dectin)-2 to mucins is mediated by the high mannose *N*-glycans present on mucins.[82] This highlights the importance of mucin *N*-glycosylation profile in modulating host innate immunity and resolving inflammation.

Therefore, genetics, environmental, and immune factors converge to regulate the expression of *N*-glycans that, in turn, fine-tune gut immune responses associated with both homeostasis and to inflammation onset in the context of several diseases.

3.2 Intestinal *N*-glycans as shapers of gut microbiome and immune response

In the last years, several studies have emphasized the impact of intestinal glycocalyx composition in regulating homeostatic immune response and in pathological host–microbial interactions.[83,84] While variations in intestinal *O*-linked glycans have been widely recognized as a key contributor factor,[85–90] our focus is to explore the impact of variations in *N*-glycosylation on symbiotic host–microbe communication.

The physical barrier provided by the epithelium is enhanced by a complex and diverse layer of *N*-glycans that are attached to the cell surface – the mucosa glycocalyx.[84] One of the major aspects of intestinal *N*-glycans is their dual role: acting as an additional shield that protects host cells from intestinal pathogens and driving the composition of the gut microbiota.[91]

A key example is the expression of intestinal epithelial α1,2-fucose that can act as an essential player in defending against several pathogenic and opportunistic microorganisms, by modulating their attachment to the epithelium.[92–94] Indeed, *Fut2*-deficient mice display increased susceptibility to infection by *S. typhimurium* and *Candida albicans*, underscoring

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the protective role of gut epithelial α1,2-fucose against some pathogens.[95,96] The expression of epithelial α1,2-fucose was also shown to promote the growth of commensal microbes such as *Porphyromonadaceae, Ruminococcaceae,* and *Bacteroides* species which, in turn, confers resistance to pathogenic and opportunistic microorganisms.[97,98] Additionally, *Fut2* mutations have been associated with a dysbiotic gut microbiota, characterized by a reduction of *Bifidobacteria* and an expansion of the Lachnospiraceae family.[95] It is noteworthy that dysbiosis has been tightly associated with several diseases, including IBD and colon cancer.[99–103] Nevertheless, FUT2 enzyme activity have been also associated with increased susceptibility to infection by *Helicobacter pylori*, Norovirus and Rotavirus A.[104–106] This is mainly due to the capacity of these microorganisms to recognize and use sialic acid residues on the mucosal epithelia of digestive tracts as binding ligands.[107,108] Similarly, recent studies identified GlcNAc-containing glycans as potential targets for pathogenic *S. typhimurium* during invasion and colonization of the SI.[109] This demonstrates the divergent role of glycans in the interaction with intestinal microbes along the gastrointestinal tract. Together, *FUT2*-mediated fucosylation seems to be essential for the fine-tuning of microorganism regulation and survival.

N-glycans are used as a source of energy and nutrients by the gut microbiota *via* a glycan-utilization machinery, known as polysaccharide utilization loci (PULs), comprising GBPs, transporters, and glycan-degrading enzymes (CAZymes), enabling the recognition, import and degradation of different glycans.[110] The intestinal glycan degradation by commensal bacteria results in the production of short-chain fatty acids (SCFAs), such as butyrate, propionate, and acetate. SCFAs play diverse roles, being widely recognized for regulating the function of the immune compartment reinforcing the pivotal role of glycans in the symbiotic crosstalk between host and microbiota. Particularly, butyrate is an important energy source of colonocytes, exhibiting both anti-inflammatory and anti-tumor properties.[111–113] Butyrate strongly induces Treg differentiation and promotes IL-22 release from $CD4^+$ T cells and innate lymphoid cells (ILCs).[114,115] Additionally, this microbial metabolite also exhibits an antimicrobial effect against *Acinetobacter baumannii, E. coli,* and *Staphylococcus pseudintermedius*, inducing the depolarization of the bacterial membrane and acidification of the cytosol.[116,117] Furthermore, butyrate has also been associated with suppressed growth and increased apoptosis in cancer cells.[118] Of note, different studies demonstrated that butyrate sensing and signaling are altered in both IBD and colorectal cancer (CRC) patients.[119,120]

Gut microorganisms differ in the number and type of glycans that they can utilize.[121] While certain microbes, such as *Bacteroidota*, are considered "generalists", *i.e.*, degrading different types of glycans, other species, including members of the *Firmicutes* phylum, are considered more "specialist" being restricted to one or few glycans.[121,122] Some *Bifidobacterium* species were shown to have the ability to remove *N*-glycans from glycoproteins *via* β-mannosidases, allowing its rapid and selective growth.[123] Importantly, *Phascolarctobacterium spp*., a specific

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population of the gut commensal bacteria, utilize host-derived GlcNAc for their own growth.[124] The abundance of *Phascolarctobacterium spp*. is particularly relevant since it restricts the opportunistic growth of *C. difficile*, which is responsible for one of the most common clinical complications in IBD patients.[125] Indeed, the degradation of GlcNAc impacts the virulence and pathogenesis of several human opportunistic pathogens, including *C. albicans, E. coli, Vibrio cholerae, Leishmania donovani*, and *Mycobacterium*.[126]

Bacteroides thetaiotaomicron, a predominant bacterial species in the ileum and colon of humans and mice, is equipped with a particular PUL to access mammalian high-mannose *N*-glycan, yeast α-mannan and complex *N*-glycan from human Ig.[127] In particular, B. *thetaiotaomicron* metabolized complex *N*-glycans *via* cleavage of GlcNAc-b(1,2)-mannose linkages,[128] and utilized epithelial α1,2-fucose as a nutrient source.[129] The degradation of fucose residues controlled the expression of several bacterial genes, determining the fitness and behavior of bacteria.[130] Furthermore, upon incorporation and metabolization of epithelial fucoses, *Bacteroides fragilis* synthetized fucosylated capsular polysaccharides, which are components of bacterial outer membrane.[129] The incorporation of host glycans by luminal microbes in their own glycoconjugates is an important strategy of colonization, acting as a mask to avoid the detection by immune cells.[128] This type of phenomenon calls for a seminal concept named as molecular mimicry.[131,132] Moreover, molecular mimicry concerns as well, when alterations in *N*-glycans confounds the immune system and are recognized as non-self-glycan structures, ultimately promoting the loss of immunological self-tolerance.

Recent evidence reported the ability of gut microbiota to reprogram intestinal epithelial glycosylation.[68] Unlike conventionally housed mice, germ-free mice revealed impaired expression of α1,2-fucose in the ileum. Accordingly, the colonization of germ-free mice with commensal *B. thetaiotaomicron* restored the expression of *Fut2* and α-1,2-fucosylation.[97,98] Likewise, *B. fragilis* and segmented filamentous bacteria (SFB) were reported as inducers of epithelial α1,2-fucose expression.[78] In fact, SFB-deficient mice showed impaired ileal epithelial fucosylation, suggesting that SFB is a major microbial stimulus inducing epithelial α1,2-fucosylation in homeostasis.[130] Importantly, commensal bacteria also promote intestinal epithelial fucosylation by an indirect mechanism involving mucosal immune cells. Goto and colleagues have shown that commensal gut microbiota interacts with ILCs, specifically ILC3, stimulating the production of IL-22, which in turn induces the upregulation of *Fut2* expression on IECs. In addition, the same group reported that constitutively production of lymphotoxin by ILC3s plays a similar role in *Fut2* induction.[78,92]

Collectively, due to their crucial role in integrating microbial and immunological cues, *N*-glycans emerged as a key driving factor of the pathologic events, leading to development of immune-mediated diseases.

3.3 Glycans as modulators of immune-mediated responses in chronic inflammation

Together with inflammation-induced alterations in epithelial glycocalyx and its role in gut microbiota, modulation of *N*-glycans have a direct role

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on immune cells. Herein, we will highlight how changes in *N*-glycans expression shape the immune-mediated response during inflammatory conditions.

T cell hyperactivation is central to the pathology of chronic inflammation and IBD. As abovementioned, glycans can mediate T cell function, activity, survival, migration and cytokine production through the regulation of different key receptors.[26] Our group demonstrated that UC patients present a deficiency in β1,6-GlcNAc–branched *N*-glycans on colonic $CD4^+$ T cells, associated with T cell hyperactivation and disease severity.[8] At the genetic level, we revealed that UC patients display a *MGAT5* single nucleotide polymorphisms (SNPs), associated with reduced expression of *MGAT5*-coded enzyme (GnT-V).[11] We further demonstrated that the hyperactivation of colonic T cells in UC patients can be controlled through the *ex vivo* metabolic supplementation with GlcNAc. Additionally, we showed that *Mgat5* knockout (KO) mice present higher disease severity to DSS-induced colitis, which was attenuated when supplemented with GlcNAc.[9] Interestingly, mucosal T cells from inflamed tissue of IBD patients, as well as from mice under colitis, revealed increased levels of core fucosylation. Indeed, *Fut8* KO mice produced lower levels of Th1 and Th2 cytokines, leading to the development of milder colitis symptoms. Consistently, core fucosylation of the TCR is required for T cell signaling and production of inflammatory cytokines.[47] Furthermore, differential glycosylation of Th1, Th2 and Tregs can modulate their fate through binding to galectins. Th2 and Treg cells express higher levels of α2,6-sialylation compared to Th1 cells, which protects them from Gal-1-mediated apoptosis.[133] Notably, it has been reported that UC patients display elevated levels of α2,6-sialylation in colonic tissue and that the deletion of *ST6GAL1* gene inhibits the proliferation and activation of $CD4^+$ T cells, leading to a reduction in pro-inflammatory cytokines.[134] Morosi and colleagues described that the upregulation of *ST6GAL1*, detected in colonic biopsies of IBD patients, masks T cells from Gal-1, inhibiting its binding and thereby inducing inflammation.[135] Particularly, Gal-1 expression was found to be upregulated in inflamed tissue of IBD patients compared to uninflamed tissue and healthy controls.[136] Conversely, Gal-2, Gal-3, Gal-4 and Gal-9 expression were downregulated in inflamed areas.[36,137] Interestingly, the expression pattern of galectin-encoding genes of IBD patients in remission was similar to the one observed in healthy controls,[136] suggesting that the resolution of inflammation leads to the recovery of normal galectin gene expression.[71]

Even though B cells play a pivotal role in gut immunity, there is still a gap of knowledge regarding the impact of B cell *N*-glycosylation in the regulation of homeostasis and inflammation. Host intestinal mucosa contains the largest population of plasma cells, displaying 80% of all antibody-secreting B cells.[138] Alterations in the glycosylation of IgG have demonstrated to impact the biological functions and structure of IgG, affecting their anti-inflammatory and pro-inflammatory phenotype.[11,139–142] In fact, it was described that patients with IBD exhibit lower levels of IgG galactosylation, as well as sialylation, which was

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associated with increased severity of the clinical outcomes.[139,140,143] Terminal sialic acids were shown to display an anti-inflammatory function on IgG, by increasing expression of the inhibitory FcγRIIB and reducing affinity to activating FcγRIII/IV,[144,145] which suggests that sialic acids in Fc promote a conformational change in IgG. Additionally, terminal galactose was shown to increase affinity of IgG to inhibitory FcγRIIB and C-type lectin–like receptor dectin-1, promoting an anti-inflammatory phenotype.[146] In IBD patients, a galactosylated IgG with terminal GlcNAc was shown to enhance antibody-dependent phagocytosis.[147] The effector function of a galactosylated IgGs might be explained either by the bind of mannan-binding lectin (MBL) selectively activating the lectin complement pathway,[148] or by the dependence on activating Fc receptors.[149] This decrease in IgG galactose can be related to the downregulation of the *β4GalT* gene (that encodes for β-1,4-galactosyltransferase (β4GALT1) enzyme, which adds galactose on top of GlcNAc residues) in B cells from CD patients.[139] Additionally, specific *MGAT5* gene SNPs observed in UC patients were found to be associated with agalactosylation of IgGs.[11] UC patients also display decreased fucosylation and increased bisecting GlcNAc on IgGs.[140] Both core fucosylation and bisecting of the Fc region are key determinants of antibody-dependent cell-mediated cytotoxicity (ADCC).[150–152] In fact, non-fucosylated IgG1 showed a 100-fold increase of ADCC compared to its fucosylated counterpart.[151]

While most of the studies on the role of glycans in chronic inflammation and IBD have focused on adaptive immunity, it is important to note that innate immune cells also rely on *N*-glycans to modulate their function and intercellular communication. For instance, in macrophages, core fucosylation of CD14 is essential for Toll-like receptor (TLR)4 and TLR2 signaling. Despite the wealth of knowledge on the *N*-glycosylation role in adaptative immunity, *N*-glycans significance in innate immune cells during chronic inflammation remains largely unexplored, opening a new avenue of research to explore in the field.

These findings collectively underscore the critical role of immune cell *N*-glycosylation in regulating the immune response during chronic inflammation, acting as a key player in both health and disease.

3.4 Cellular glycosylation in the breakdown of self-tolerance in autoimmunity

Glycosylation is a tissue and cell specific process fundamental to defining the identity of a cell or an organism. However, changes in the *N*-glycosylation profile can lead to abnormal recognition of "self" and "non-self" antigens associated with autoreactive immune responses. This concept known as "glycan mimicry" is at the basis of the loss of self-tolerance associated with autoimmunity.[153]

Autoimmune diseases are chronic and debilitating diseases characterized by the dysregulation of the immune system, resulting in the uncontrolled immune response targeting its own tissues within the body and resulting in the damage of different organs. In this section, we will

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describe how changes in glycocalyx composition contribute to the immunopathogenesis of several autoimmune diseases (see Fig. 2).

3.4.1 Impact of altered glycocalyx in autoimmune diseases. A classic prototype of autoimmune disease is Systemic Lupus Erythematosus (SLE), a multisystemic disease characterized by autoreactive immune cells and abnormal autoantibody production against self-antigens which, together, lead to the deposition of immune complexes and organ damage. One of the most severe complications of this disease is lupus nephritis (LN), which involves renal damage that ultimately can lead to organ dysfunction and the need for kidney transplantation. In this disease, a full glycome characterization of LN tissue revealed a unique glycan repertoire of renal epithelial cells.[13] In fact, imaging mass spectrometry analysis disclosed an accumulation of mannose-enriched *N*-glycans in renal epithelial cells from LN patients that were not observed in healthy kidney or other kidney diseases' samples. The accumulation of less complex *N*-glycans was related to the downregulation of Mannosidase Alpha Class 2A Member 1 (*MAN2A1*) gene (encoding α-mannosidase-II enzyme) and upregulation of Protein O-Mannosyltransferase 1 (*POMT1*) gene (encoding mannosyltransferase enzyme) expression. This was associated with increased infiltration of a subset of immune cells[12] that express DC-SIGN, which recognizes mannosylated glycans. Importantly, this specific glycosignature has shown predictive value, as it stratifies patients according to the risk of developing chronic kidney disease (with a specificity of 93%), being a potential diagnostic and prognostic glyco-biomarker to be used in the clinic. In accordance, a case-report study on two sisters with α-mannosidosis diseases showed an association with a truncated *N*-glycome (accumulation of mannose-enriched *N*-glycans) and the development of SLE disorder.[154] Mice studies have also pinpointed glycans as important modulators of tissue homeostasis. The *MRL-lpr* lupus mouse model demonstrated an increased accumulation of less-complex *N*-glycans in kidney proteins, including paucimannose and mannosylated-type *N*-glycans.[155] The *in vivo* deficiency of α-mannosidase-II (enzyme that hydrolyzes a mannose residue) resulted in the accumulation of mannose- and hybrid-type glycans. This resulted in the development of a disorder that resembles SLE disease, with glomerulonephritis, accumulation of autoantibodies and C3 complement deposition, highlighting the importance of less-complex *N*-glycans in the immunopathogenesis of this disease.[24,156] Similarly, glycoengineered *Mgat5*-KO mice showed increased severity of colitis and experimental autoimmune encephalomyelitis (EAE), as well as spontaneous lupus-like syndrome with aging.[9,12] In rheumatoid arthritis (RA) the pro-inflammatory milieu around the joints contributes to the destruction of bone and cartilage, leading to severe pain and joint swelling. Notably, in inflamed joints of RA patients, it was observed a downregulation of *ST6Gal1* expression of synovial fibroblasts in tumor necrosis factor (TNF)-dependent manner, which consequently reduced the content of α2,6-sialylated glycans and potentiated the inflammation and disease.[157] Moreover, this

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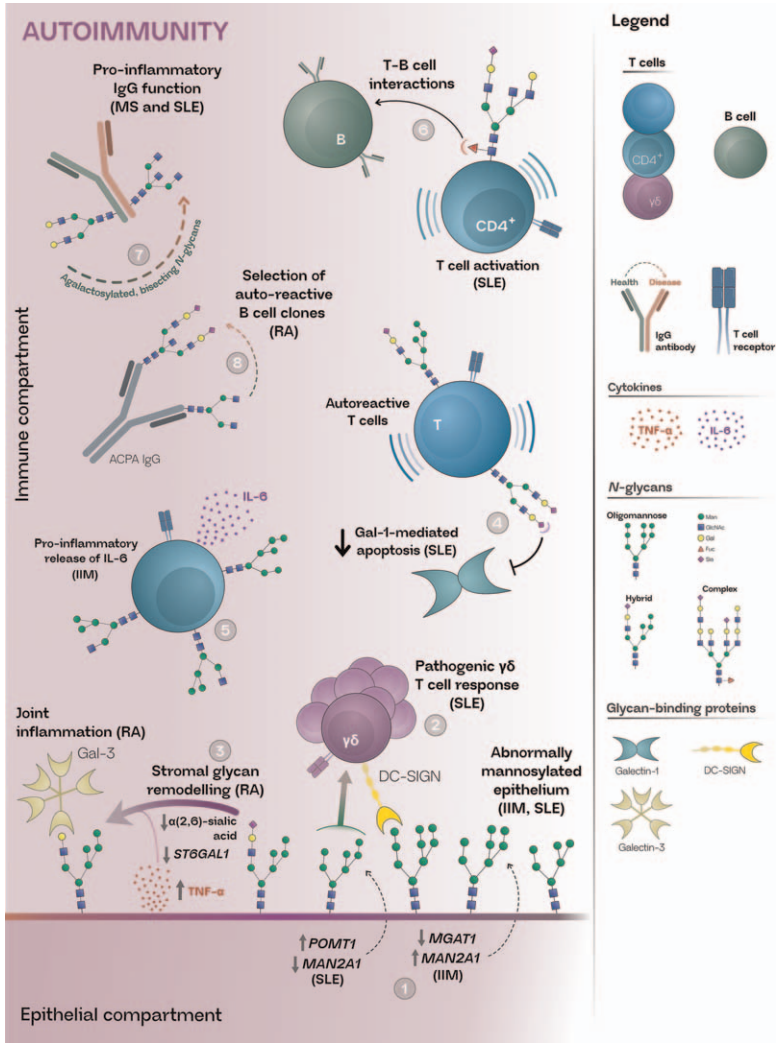


Fig. 2 Abnormal cellular *N*-glycosylation as a mechanism involved in the breakdown of self-tolerance and in shaping autoimmune-mediated disorders. *N*-glycosylation is fundamental for the "self" and "non-self" discrimination, although changes in *N*-glycans profile can results in the loss of self-tolerance associated with autoimmunity. (1) Abnormal expression of glycogenes (*POMT1, MAN2A1* in SLE and *MGAT1, MAN2A1* in IIM) involved in the early steps of *N*-glycosylation pathway, leads to the accumulation of mannose-enriched *N*-glycans at the epithelial cell compartment from SLE (kidney) and IIM (muscle) disorders. (2) The presence of less-complex *N*-glycans in renal epithelial cells from SLE patients, resulted in increased infiltration of pathogenic DC-SIGN-expressing γδT cells. (3) Pro-inflammatory environment of synovial tissue from RA patients is characterized by higher levels of TNF-α, resulting in decreased α2,6-sialylated glycans (downregulation of ST6Gal1) on synovial fibroblasts and Gal-3-mediated inflammation. (4) In SLE, autoreactive T cells display an abnormal glycosylation profile (increased terminal sialylation) and escape to the Gal-1-mediated apoptosis. (5) The absence of complex *N*-glycans on $CD4^+$ T cells from IIM patients, results in increased pro-inflammatory phenotype, with the secretion of IL-6 cytokine. (6) The higher levels of core fucosylation on $CD4^+$ T cells from SLE patients associates with higher T cell activation, T and B cell interaction and disease severity. (7) In SLE and MS, the IgG Fc glycosylation display reduced galactosylation and elevated bisecting GlcNac structures, which associates with increased IgG effector function. (8) The ACPA-Fab region from RA patients, showed high levels of bisecting GlcNac, galactose and sialic acid content, resulting in the selection of autoreactive B cells.

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inflammatory environment promoted the expression of β1,4-galactosyltransferase-I in synovial tissue, suggesting a pathogenic role of inflammation and stromal *N*-glycans in RA immunopathogenesis.[158] Additionally, FUT1 enzyme was found to be upregulated in synovial fibroblasts of RA patients, resulting in increased production of fucosylated glycans, associated with fibroblast proliferation and adhesion, contributing to the tissue damage and inflammation of RA disease.[159] Similar alteration was observed in Psoriasis (PsO), an inflammatory autoimmune skin disease characterized by hyperproliferation of epidermal keratinocytes, where some patients may also develop joint damage known as psoriatic arthritis (PsA). The *FUT8* gene was found to be overexpressed on skin biopsies of psoriatic patients and associated with disease severity. This glycosylation alteration led to increased activation of EGFR protein kinase and was related to increased proliferation of epidermal keratinocyte cells, emphasizing the impact of altered glycans during the development of psoriasis.[160] Idiopathic inflammatory myopathies (IIM) are a group of rare heterogeneous disorders that affect the muscles and skin, leading to muscle weakness, atrophy, and ultimately imposing a poor quality of life on patients. A recent study characterized the glycosylation profile of muscle biopsies from IIM patients and found increased levels of high-mannose *N*-glycans due to decreased *MGAT1* and increased *MAN2A1* gene expression.[14] This unique tissue glycome was associated with poor prognosis of IIM patients. Another subgroup of myopathies, the GNE autosomal genetic myopathy (caused by mutations in the GNE gene), resulted in the loss of sialic acid synthesis in muscle cells and in misfolding of some glycoproteins.[161] Interestingly, evidence suggested that oral glycan supplementation promoted branching *N*-glycosylation by glycoreprogramming the muscle environment, recovering the homeostatic *N*-glycans.[14,162] Type-1 diabetes mellitus (T1DM) is a subgroup of metabolic disorders characterized by autoimmune pancreatic β-cell destruction resulting in low insulin levels and hyperglycemia. Non-secretors individuals with a non-functional *FUT2* allele and unable to express blood group antigens in secretions and on mucosal surfaces are associated with increased susceptibility to T1DM and resistance to certain host infections.[81]

Altogether, these evidences suggest that cellular glycome, and specifically changes in *N*-glycans may act as potential danger signals (DAMPs) for the immune system, contributing to the immunopathogenesis of autoimmune diseases.

3.4.2 Immunomodulatory role of glycans in autoimmunity: glycans as shapers of cellular and humoral immune responses. Abnormal glycosylation of the immune counterparts is also widely associated with a pathogenic autoreactive phenotype and the hyperactivation of the immune system.

Alterations in the *Mgat5* gene expression are well described in the context of T cell regulation and autoimmunity.[8,163] Demetriou and colleagues have reported that the absence of β1,6-complex *N*-glycans affects the cell surface expression of immune checkpoint receptors

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[programmed cell death 1 (PD-1) and cytotoxic T-lymphocyte antigen 4 (CTLA-4)] and results in TCR clustering, T cell hyperactivation, reduced galectin–glycoprotein lattice and susceptibility to autoimmunity.[163] In accordance, in multiple sclerosis (MS), a chronic autoimmune disease characterized by demyelination and neurodegeneration of the central nervous system, the peripheral blood mononuclear cells (PBMCs) from these patients present a downregulation of the GnT-V enzyme activity.[164] Consistently, the conditional Lck-Cre^{+}Mgat2$^{fl/fl}$ mice model, where T cells lack tri- and tetra-antennary branched *N*-glycans, developed an increased susceptibility to EAE a model that mimics MS. In fact, this susceptibility was potentiated in mice treated with kifunensine, promoting the accumulation of mannosylated glycans and inhibiting the compensatory poly-LacNAc structures.[165] Moreover, the lack of Gal-3 was shown to attenuate EAE disease severity, where Gal-3 KO DCs produced IL-10 and inhibited the release of pro-inflammatory cytokines by T cells.[166] On the other hand, the absence of Gal-1 in MS-like disease resulted in Th1 and Th17 pro-inflammatory responses and microglial activation. Notably, the administration of Gal-1 suppressed this phenotype and attenuated EAE disease.[167] In addition, the absence of fucose attached to the *N*-linked glycans in *Fut8*-KO mice resulted in less CD4^{+} T cell activation, controlled T and B cell interaction, attenuating EAE syndrome.[46] Moreover, specific *MGAT1 and MGAT5* SNPs, together with CTLA-4, IL-2 receptor-α (IL2RA) and IL-7 receptor-α (IL7RA) SNPs, were correlated with an increased risk and severity for developing MS and T1DM.[168–170] Genome-wide analysis of PsO patients revealed that specific *MGAT5* SNPs were also associated with altered *N*-glycosylation profile of activated T cells from these patients.[171] Consistently, the conditional *Rag1^{Cre}Mgat1$^{fl/fl}$* glycoengineered mice model, harboring high-mannose *N*-glycans only in lymphocytes, developed spontaneous inflammation in colon and kidney tissues, as well as imiquimod-induced psoriasis.[30] In SLE, resting T cells from patients exhibited an activated-like phenotype according to the glycosylation status as well as reduced levels of Gal-1 binding.[172] In fact, it was a consequence of increased expression of ST6GAL1/NEU1 and ST3GAL6/NEU1 ratios by the cells, resulting in the escape of autoreactive T cells by the Gal-1 apoptotic effect. Interestingly, serological levels of Gal-9 were significantly increased in patients with SLE and correlated with disease activity. This association was particularly pronounced in patients with renal damage, suggesting that Gal-9 could serve as a serologic marker of SLE pathogenesis.[173] In aged *Mgat5*-null mice and α-manosidase-II deficient mice (both develop SLE-like disease spontaneously), it was described, respectively, the infiltration of DC-SIGN–expressing γδ T cells and the macrophage mannose receptor (MMR) protein expression on mesangial cells and macrophages. This unlocks a new axis of glycans and glycan-binding proteins in SLE immunopathogenesis.[12,24] Evidence have shown that DC-SIGN is overexpressed in the podocytes of LN patients, and mice treated with anti-DC-SIGN antibody hampered the autoimmune signs by reducing proteinuria levels, T cell proliferation, proinflammatory cytokines, and tissue damage.[174] In some autoimmune diseases, the *FUT8* gene is described to

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affect T cell activity and response, particularly for SLE patients in which increased levels of core fucosylation of $CD4^+$ T cells are associated with disease severity.[46] Myasthenia Gravis (MG) is a neuromuscular autoimmune disorder characterized by the weakness of the muscles and impaired neuromuscular transmission. In thymic epithelial cells from MG patients, was described that increased expression of Gal-1 mediates the interaction of thymic epithelium with T cells, leading to abnormal thymus morphology and disease.[175] In addition, aged mice lacking Gal-1 (*Lgals1*$^{-/-}$) developed a spontaneous inflammation of salivary glands, similar to Sjögren's syndrome (SjS), an autoimmune disease that affects exocrine glands.[176] Likewise, this phenotype was also observed in other mice-induced autoimmune-like diseases and linked with tissue damage.[133,177] Similarly, in IIM, the GNE activity promoted the poly-LacNAc extension, affecting the binding of Gal-1, which in turn might disturb T cell signaling and activity.[161] In IIM patients, it was also reported that the absence of branched *N*-glycans in $CD4^+$ T cells was associated with an increased pro-inflammatory phenotype and the release of IL-6 cytokine.[14] Opposingly, Gal-3 KO mice showed attenuated symptoms of antigen-induced arthritis, a mouse model mimicking RA disease, with less joint inflammation and decreased levels of autoantibodies and proinflammatory cytokines.[178]

The role of B cells and IgGs cannot be left behind in the context of autoimmunity, where autoantibodies are one of the hallmarks of this group of diseases. Changes in B cells and IgG glycome have been described in different disease contexts, from autoimmune and inflammatory diseases to infections and cancer.[179] A recent study described a dose-dependent and dual role of *N*-glycan branching in B cells function. Here, lack of *N*-glycan branching on B cells resulted in reduced CD19 and BCR surface retention and promoted TLR signaling, generating a pro-inflammatory environment, that together exacerbated inflammatory demyelination in EAE mice model.[180] Furthermore, the interplay of Gal-9 with BCR controls its signaling transduction. Consistently, lower BCR thresholds and an accelerated rate of spontaneous autoimmunity with renal damage and autoantibody production were observed in aged Gal-9-deficient mice.[181] In MS patients, the analysis of IgG Fc glycosylation in the cerebrospinal fluid revealed reduced afucosylation and galactosylation, concomitantly with elevated bisecting GlcNAc in *N*-glycans associated with increased IgG effector functions. Moreover, the plasma IgG of MS patients displayed lower core fucose and increased content of high-mannose Fc *N*-glycans, which was also associated with their pro-inflammatory potential. In SLE, the Fc IgGs presented decreased galactosylation, sialylation, fucosylation along with increased bisecting GlcNAc, suggesting decreased immunosuppressive potential of circulating IgGs.[182,183] Lower levels of IgG Fc galactosylation were also described in MG, PsA and IIM patients.[183–185] Similarly, in RA disease, the first report of altered IgG-Fc glycome in patients described elevated agalactosylated glycans content compared with healthy individuals, which was further validated by others and associated with RA disease activity.[186,187] Moreover, in RA-specific anti-glycan antibodies against

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citrullinated peptides (ACPA-IgGs), the ACPA-IgG1-Fc glycosylation of synovial fluid differs from total serum IgG1, with reduced sialylation and elevated agalactosylation of ACPA-Fc on synovial fluid.[188] In addition, the lack of galactose in ACPA-IgG-Fc is associated with RA disease progression and inflammation.[64] Interestingly, the human IgG Fab regions can also carry *N*-glycosylation consensus sequences, and the Fab component of RA patients displays high levels of bisecting GlcNAc, galactose and disialylation, which may also be reflected in antigen-binding capacity.[189] Importantly, the presence of *N*-glycosylation sites in ACPA-IgG Fab affects B cell antigen binding and BCR signaling, contributing to the selection of autoreactive B cell clones, that might be involved in the RA pathogenesis.[190,191] In MG disease, specific autoantibodies targeting crucial molecules at the neuromuscular transmission system, such as the nicotinic acetylcholine receptor (AChR) and muscle-specific kinase (MuSK), display elevated *N*-linked glycosylation motifs in the variable domains of AChR and MuSK autoantibodies associated with MG disease when compared with healthy controls.[183,192]

Despite the differences in IgG glycome (Fc- and Fab-dependent) associated with pro-inflammatory responses in autoimmunity, the cellular and molecular mechanisms underlying this response remain to be elucidated, and further studies are needed to fully characterize these mechanisms.

3.5 *N*-Glycosylation in the tumor microenvironment

Changes in cellular glycosylation have been considered a hallmark of cancer, mediating critical events of tumor progression. In fact, glycans play a pivotal role in each pathophysiological step of malignant transformation and particularly in cancer immunoediting, promoting tumor immune evasion.[6]

Here, we will discuss the multiple roles of *N*-glycans in the tumor microenvironment (TME), either as major regulatory molecules of cancer cell behavior but also as master regulators of immune cell functions in the TME, such as innate immune cells and tumor infiltrating T cells (see Fig. 3).

3.5.1 Tumor cell glycosylation alterations: the enemy at the dynamic crosstalk between cancer cells and immune cells. Tumor cells acquire a variety of mechanisms to evade the immune system, and one of them is the expression of aberrant *N*-glycosylation at the cell surface. In the present section, we will describe some mechanisms related to *N*-glycosylation that tumors use to hijack the immune system.

The most remarkable alteration described in tumor cells is the abnormal expression of sialic acids. This coat of sialoglycans works as a shield, protecting tumors from being recognized and eliminated by immune cells, particularly through Siglecs.[193] In hepatocellular carcinoma (HCC) tumor tissues, a strong inflammatory response was observed with lower expression of 2,6-sialic acid.[194] In accordance, in an HCC cell line, it was described that overexpression of *ST6Gal1* decreased $CD8^+$ T cell infiltration in TME, as well as their cytokine production, namely IFNγ

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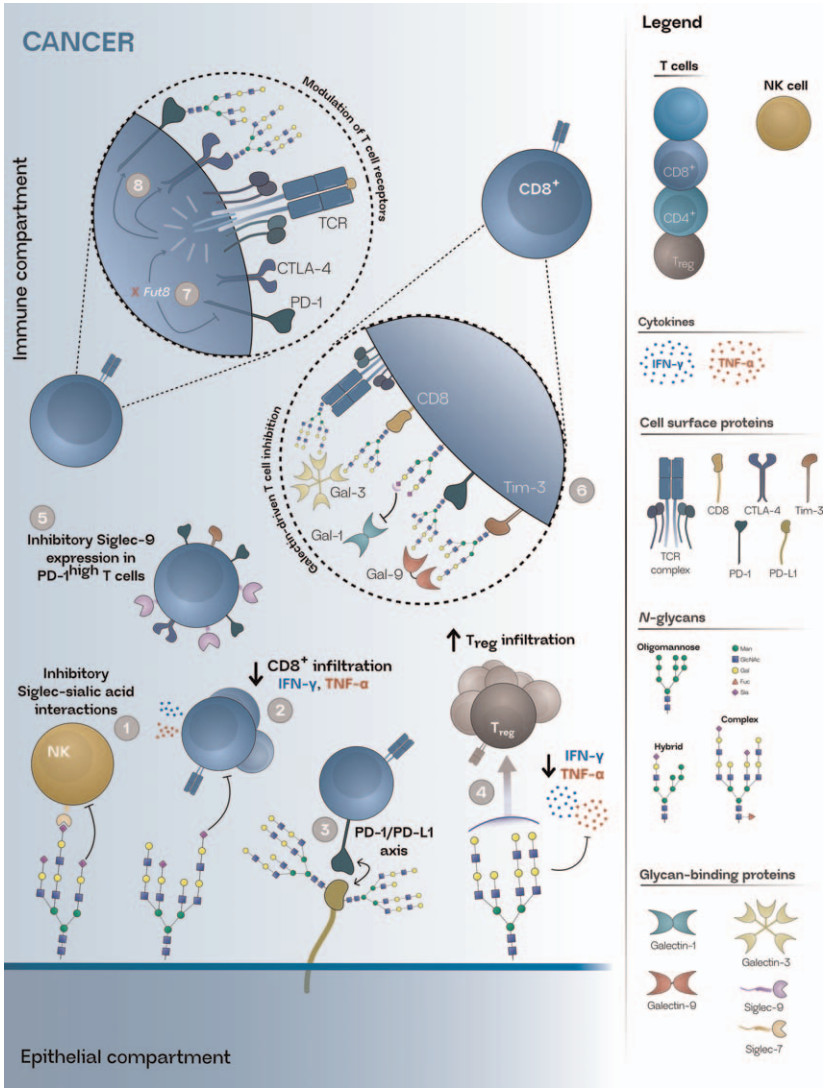


Fig. 3 Glycans as key regulators of immunological networks in cancer progression. In cancer, overexpression of branched *N*-glycans display a dual-edged sword, orchestrating tumor immune evasion. Branched *N*-glycans not only impose invasive and pro-malignant features to cancer cells but also serve as camouflage, shielding tumor cells from immune surveillance, while simultaneously disrupting immune effector functions. (1) Sialic acids present on tumors inhibited NK cells activation by engaging with Siglec-7 and -9. (2) Overexpression of sialic acids, through upregulation of *ST6Gal1*, decreased $CD8^+$ T cell infiltration in tumor microenvironment (TME), as well as the production of IFNγ and TNFα. (3) The interaction of PD1/PD-L1 is coordinated by *N*-glycosylation. (4) Expression of β1,6-GlcNAc branched *N*-glycans in cancer cells dampened the release of cytokines such as IFNγ and TNFα and correlated with increased Treg infiltration in the TME. (5) Siglec-9 is highly expressed on PD-1^{high} tumor infiltrated T cells, associated with the expression of inhibitory receptors and inhibition of T cell anti-tumor response. (6) Altered *N*-glycans on T cells surface regulate anti-tumor immune response through interaction with GBPs, such as galectins, dampening their immune function and perpetuating immunosuppression within TME. (7) *N*-glycosylation also regulates the expression of PD1 at T cell surface and upon FUT8 inhibition PD1 expression is reduced, leading to T cell activation. (8) TCR activation leads to increase of branched *N*-glycans in CTLA-4, resulting in T cell suppression.

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and TNFα.[195] Additionally, sialic acids present on tumors inhibited NK cells activation by engaging with Siglec-7 and -9.[196,197] In line with these findings, in fibrosarcoma cell lines, tumor cell desialylation led to the induction of NK cell cytotoxicity.[198] Moreover, a decrease in Siglec-9 ligands on tumor cells significantly reduced immunosuppression by enhancing T cell effector functions and their anti-tumor capacity.[199,200] The key impact of sialylation is also demonstrated by the role of CD24. CD24 is a heavily sialylated glycoprotein and tumors expressing CD24 were shown to preclude phagocytosis by tumor-associated macrophages (TAM) through Siglec-10 binding.[201] On the other hand, Gal-1 interacts with glycans in the TME, mediating malignant processes. In fact, α2,6-sialic acids, highly expressed on tumor cells, prevented Gal-1 binding, which protected them from Gal-1-induced death.[202,203] However, Gal-1 expression by tumors promoted the presence of TAMs harboring immune checkpoints, such as PD-L1 in the TME.[203] Gal-1 role in the TME and particularly on T cells will be further discussed in more detail in the section *T cell surface glycosylation dynamics during tumor progression*.

The aberrant overexpression of β1,6-GlcNAc branched *N*-glycans structures has been widely described on tumor cells. In fact, overexpression of complex branched *N*-glycans was described to induce pro-malignant and pro-invasive phenotypes, associated with poor prognosis of gastrointestinal cancer patients.[204–207] Recently, our group had shown that the aberrant expression of β1,6-GlcNAc branched *N*-glycans in CRC acts as an important immune checkpoint in cancer immunosurveillance and immunoediting. These glycan structures dampened the release of cytokines such as IFNγ and TNFα and correlated with increased Treg infiltration in the TME. Besides, the inhibition of β1,6-GlcNAc branched *N*-glycans enhanced immune recognition of cancer cells.[18] In CRC tissues, a downregulation of bisecting *N*-glycans was observed to be associated with IFNγ resistance of tumor cells. Additionally, *MGAT3* overexpression was able to restore IFNγ sensitivity of tumor cells.[208] More recently, the expression of B3GNT2, involved in the biosynthesis of poly-*N*-acetyllactosamine chains on tumors was described to decrease T cell activation by disrupting tumor-T cell interactions. Moreover, treating tumor cells overexpressing B3GNT2 with kifunensine was able to restore T cell function, increasing IFNγ production.[209] Altogether, these highlighted the impact of different *N*-glycan structures in tumor immune evasion.

One of the best described immunosuppressive pathways in cancer is the PD-1/ programmed cell death 1 ligand 1 (PD-L1) axis. In this axis, the glycosylation of PD-L1 plays a major role in determining how it binds to its receptor, PD-1.[210–212] In fact, β-1,3-*N*-acetylglucosaminyl transferase (B3GNT3) expression was described to be essential for the interaction of PD-L1 with PD-1. Additionally, downregulation of B3GNT3 led to improved cytotoxic anti-tumor T cell responses.[213]

Tumor cells produce antigens that can trigger an immune response. These tumor antigens can also be *N*-glycosylated, interfering with their recognition by the immune system. In fact, when the tumor antigen tyrosinase has its *N*-glycan chain at Asn317 site mutated, its presentation

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by MHC-I is increased.[214] Interestingly, sialylated antigens bind to Siglec-E, mouse homolog of Siglec-9 in humans, on mouse DCs, which favored the differentiation of antigen-specific Tregs and reduced effector T cell numbers.[49]

Overall, these evidences promptly suggest the importance of altered *N*-glycosylation in the defense of cancer cells against anti-tumor immune attack, giving tumors an advantage to develop and proliferate.

3.5.2 GBP interactions as regulators of innate immune responses in tumor microenvironment. As previously stated in this chapter, GBPs are an essential part of the immune response, being commonly expressed by innate immune cells.[215] In this section, we are going to focus on CTLs and their role in cancer upon *N*-glycan recognition.

MBL can bind Le^A and Le^B in *N*-glycans expressed by CRC cell lines, specifically in the structures that decorate CD26.[216,217] Indeed, MBL-mediated cellular cytotoxicity suppressed tumor growth in athymic mice injected subcutaneously with a human CRC cell line.[218] The gastric cancer cell line MKN45 displayed high DC-SIGN binding motifs expression, and the inhibition of DC-SIGN and MR in monocyte-derived DCs (moDCs) led to lower CD86 levels when co-cultured with MKN45 cells.18 In addition, several human tumor cell lines and cancer tissue samples displayed paucimannose and high-mannose *N*-glycans that can bind MR.219 Dectin-1 was also capable of binding *N*-linked saccharides in the B16 melanoma cell line, culminating in the activation of NK cells and induction of an anti-tumoral response220.

On the flipside, DC-SIGN had been implicated in DC immunosuppression. In CRC cell lines, Carcinoembryonic Antigen (CEA) displayed high levels of fucosylated *N*-glycans,[221] in particular, increased Le^X and Le^Y epitopes,[222] which can be recognized by DC-SIGN expressed by immature DCs, potentially facilitating immune evasion.[222] Another study had reported that DC-SIGN detected elevated Le^A/Le^B antigens in CEA from the SW1116 CRC cell line, inducing the secretion of immunosuppressive cytokines, such as IL-10[223]. Mac-2 Binding Protein (Mac-2BP) was also found to be expressed in CRC cell lines and exhibited aberrant Lewis antigen expression that was bound by DC-SIGN, an effect that was lost after PNGase F treatment, which cleaves *N*-linked glycans from glycoproteins, indicating a dependency on *N*-glycans.[224] This interaction impaired DC maturation, potentially limiting anti-tumoral activity.[224] Cancer Antigen 125 (CA-125) is a glycoprotein frequently associated with Ovarian Cancer (OC) that presented significant differences in *N*-glycosylation sites in OC patients.[225] Interestingly, the MR expressed by TAM can bind CA-125, inducing the secretion of IL-10226.

All things considered, CTLs have shown a dual role in the shaping of the TME, making them an interesting target for the development of cancer therapies.

3.5.3 T cell surface glycosylation dynamics during tumor progression. T cells have the capacity to specifically recognize and respond to tumor antigens, being crucial mediators of anti-tumor immune

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response. However, T cells are not always effective to eliminate tumor cells. Alterations to normal T cell *N*-glycosylation patterns may lead to the loss of function of these cells within the TME.[21,227,228] Here, we will depict the T cell *N*-glycosylation alterations that lead to the formation of immunosuppressive pathways in the TME, precluding an effective anti-tumor immune response.

As described in detail in the section "*Glycans as immune regulatory circuits in homeostasis*", a wide variety of T cell receptors are glycosylated,[26] and alterations in the glycosylation pattern of this subset of cells might impact the translocation, retention and signaling of these receptors on T cell surface, but also influencing its binding. Additionally, these alterations can also impact the binding to GBPs, such as galectins and Siglecs.[229] It is now established that PD-1 requires *N*-glycosylation to be retained at the T cell surface, being its *N*-glycosylation pattern altered after T cell activation towards higher expression of poly-LacNAc and core fucose. On the other hand, deficient *N*-glycosylation led to PD-1 degradation.[230] PD-1 *N*-glycosylation was also described to be essential to its binding to PD-L1, as well as for immunotherapeutic antibodies, such as nivolumab and camrelizumab.[230–232] In fact, core-fucose at the asparagine N58 residue appeared to be crucial for the action of anti-PD-1 blockade antibodies.[231,233,234] Interestingly, inhibition of Fut8 reduced PD-1 expression and improved T cell activation.[235]

Early in T cell activation, two signals are required: the first being the recognition of an antigen through MHC on APCs by the TCR and the second being a co-stimulatory signal, by molecules such as CD28.[236] Contrary to CD28, CTLA-4, also expressed on T cells, induces a co-inhibitory signal on T cells by binding with higher affinity to CD80/CD86, downregulating T cell immune response.[237] CTLA-4 is known to be regulated by glycosylation.[238,239] In fact, upon TCR activation, an increase of branched *N*-glycans in CTLA-4 occurred, leading to T cell suppression.[45,240–242] In this line, *N*-acetylglucosamine supplementation was shown to increase CTLA-4 at the T cell surface, resulting in inhibition of T cell activation.[239] Additionally, *Mgat5* KO T cells under low TCR stimulation, showed decreased CTLA-4 levels on T cell surface.[239]

Other important receptors in tumor immunosuppression were also described to be *N*-glycosylated, namely the T cell immunoreceptor with immunoglobulin and ITIM domains (TIGIT) and the Lymphocyte activation gene-3 (LAG-3).[243,244] In fact, it was shown that TIGIT PNGase F treatment prevented the binding to its receptor.[243] Curiously, LAG-3 was described to interact with Gal-3, promoting $CD8^+$ T cell dysfunction.[244] Altogether, these observations demonstrate how *N*-glycosylation of T cells receptors are a key point in the immunosuppressive processes within the TME, thus emerging as a promising target for further studies.

Galectins play an important role in the TME, mediating T cell activation.[245] For instance, Gal-1 is involved in various T cell steps, playing a role in survival, differentiation, activation and function,[246] while in cancer, it has an immunosuppressive function.[245,247] In fact, Gal-1 was correlated with decreased numbers of tumor infiltrating lymphocytes

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(TILs)[248,249] and increased Tregs in the TME,[250] promoting resistance to immunotherapy with immune checkpoint inhibitors.[248] Increased expression of α2,6-sialic acids, due to overexpression of *ST6Gal1*, on CD45 receptor was shown to prevent Gal-1 binding and consequently T cell induced death.[202] Interestingly, secretion of Gal-1 by γδ T cells promotes malignant progression, dampening anti-tumor immune response.[251] In the TME, Gal-3 abolished cytotoxic T cell function and IFN-γ secretion by establishing molecular lattices with glycans structures on the TCR and CD8.[252] Further, the use of LacNAc, a galectin ligand, to inhibit Gal-3 binding to cellular glycans, reverted anergy on human TILs.[252] Gal-9 is also known to play a part in tumor immunosuppression. In fact, T-cell immunoglobulin and mucin-domain containing-3 (Tim-3) is a receptor expressed on exhausted T cells and exerted its immunosuppressive function upon binding to Gal-9.[253] Gal-9 binds to glycans present on Tim-3, promoting apoptosis of tumor infiltrating $CD8^+$ T cells[254] and treatment with PNGase F, led to an impairment of Tim-3/Gal-9 binding.[253] Interestingly, it was recently demonstrated by Yang and colleagues that in exhausted T cells co-expressing PD-1 and Tim-3, PD-1 interacts with Tim-3 and Gal-9, attenuating Tim-3/Gal-9 induced T cell death, leading to the survival of exhausted T cells.[255] Notably, Tim-3/Gal-9 interactions on TILs were described to play a role in cancer patients' resistance to anti-PD-1 immunotherapy.[256]

Another class of GBPs that mediates immunosuppression are Siglecs, due to the aberrant expression of sialic acids on tumor cells.[257] Siglec expression on normal/resting T cells is very low, however, on TILs their expression can be significantly increased. Siglec-9 was found to be highly expressed on TILs, specifically on PD-1^{high} T cells. Additionally, these T cells presented an effector memory phenotype that expresses inhibitory receptors, such as PD-1, CTLA-4, Lag-3 and Tim-3.[199,200] Siglec-9 was found to inhibit T cell effector functions and its anti-tumor capacity.[199,200] In parallel, in mice, Siglec-E was also observed to be upregulated on TILs.[200] Furthermore, disrupting the siglec-sialoglycan interaction leads to an increase of $CD8^+$ T cells in the TME of a CRC mouse model.[200] Siglec-7 is expressed on non-exhausted effector memory $CD8^+$ T cells with increased metabolic fitness, cytotoxic capacity and cytokine production. However, upon binding to sialoglycans, Siglec-7 acts as an immune checkpoint preventing T cell function.[258] Additionally, Siglec-5 and Siglec-10 expression on T cells was also described and associated with suppression of T cell activity.[259,260]

Focusing on Treg *N*-glycosylation, it was demonstrated that the removal of *N*-glycans from the cell surface by PNGase F treatment results in decreased suppressive function, through a contact-dependent mechanism. Additionally, tri- and tetra-antennary *N*-glycans were described to be correlated with the expression of receptors involved in Treg suppression, such as CD39, CD73, inducible T-cell Co-stimulator (ICOS), Glucocorticoid-Induced TNFR Related (GITR), PD-1, PD-L1 and CTLA-4.[261] Recently, Pinioti and colleagues showed that targeting fucosylation on Tregs negatively impacts their proliferation and suppressive function, precluding tumor progression.[262] However, the full mechanism

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by which *N*-glycosylation changes Treg suppressive functions remains to be elucidated.

Lastly, on TILs, exhausted $CD8^+$ T cells were found to downregulate the oligosaccharyltransferase (OST) complex, which hampers IFNγ production.[263] Moreover, restoring OST complex decreased $CD8^+$ T cells exhaustion and increased the production of IFNγ, ultimately leading to the decrease of tumor growth.[263] Suggesting *N*-glycosylation as an important player in T cell exhaustion.

Altogether, changes in *N*-glycosylation pattern on T cells and its receptors influence T cell function within TME, highlighting the importance of T cell *N*-glycosylation in the regulation of anti-tumor immune response within TME.

3.6 Leveraging *N*-glycosylation in glycomedicine

With glycans emerging as master regulators in the pathogenesis of major diseases such as chronic inflammatory and autoimmune diseases as well as cancer, new therapeutic perspectives are rising for these sugars as potential clinical targets.[31,264,265] Glycan-based biomarkers (see Table 1) and therapies (see Table 2) have been a remarkable field of interest for the clinical management of diseases.

3.6.1 Glyco-biomarkers. Several studies have elucidated the relevance of tissue-specific glycome in various pathologies, specifically as potential glyco-biomarkers. Notably, lower levels of branched *N*-glycans in colon biopsies of IBD patients at the time of diagnosis predict non-response to standard therapy with 75% accuracy. In contrast, high

Table 1 Glycan-based biomarkers in the realms of chronic inflammation, autoimmunity and cancer.

	Biomarkers	Source
Chronic inflammation (IBD)	Low expression of complex branched *N*-glycans (correlated with high calprotectin (CRP) levels)[10]	Tissue (colon mucosa)
	Increased L-selectin, Galectin-1 and Galectin-3 levels[261–263]	Serum
	Increased levels of anti-Saccharomyces cerevisiae (ASCA), Anti-glycoprotein 2 and anti-mannobioside carbohydrate IgGs[269,270]	Serum
	Increased agalactosylated IgGs; decreased IgG fucosylation (UC) and increased IgG fucosylation (CD); higher IgG bisection (CD)[135–137,271]	Serum
Autoimmunity	High expression of mannosylated *N*-glycans[12,14] – SLE and IIM	Tissue (Kidney and Muscle)
	Increased galectin-9[168] – SLE	Serum
	Antibodies against citrullinated peptides (ACPA) IgGs variable domain glycosylation enriched in bisecting and sialylated *N*-glycan structures[273] - RA	Serum
Cancer	Increased galectin-1[264]	Serum
	Increased levels of circulating α2,6 sialylated and branched structures in plasma proteins[259]	Serum
	Enrichment in multi-antennary and terminally sialylated/ fucosylated glycans[260,265]	Tissue (colon)

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Table 2 *N*-glycosylation as potential targets for therapies in chronic inflammation, autoimmunity and cancer.

	Glycotherapy	Target	Therapeutic effect
Chronic inflammation (IBD)	GlcNAc supplementation[8,9]	T cells	Promotes *N*-glycan branching and a controlled immune response
Autoimmunity	GlcNAc supplementation[12,14,44,157] – SLE, MS, IIM	Tissue and T cells	Promotes tissue homeostasis by glycome reprogramming and controlled immune response
	Treatment with glycosidases[167,285–287] – SLE, RA	IgGs and T cells	Potentiates anti-inflammatory functions of IgGs and controlled immune response
	Administration of soluble glycosyltransferases[320] – SLE, RA	IgGs	Converts pathogenic IgGs into an anti-inflammatory phenotype
	Administration of Galectin-1[162,295] – RA, MS	T cells	Potentiates Th2-mediated anti-inflammatory responses
Cancer	Blocking galectin-1 binding[296–298]	T cells	Increased infiltration of $CD8^+$ T cells and enhanced tumor killing
	Blocking Galectin-3 binding (GM-CT-01; GR-MD-02)[299,300]	T cells	Enhances anti-tumor $CD8^+$ T cell response
	Siglec-7/9 expressing CAR-T cells[304]	Tumor cells	Promotes the recognition of tumor sialylated glycans and potentiates CAR-T-mediated anti-tumor immunity
	Treatment with β-Glucan agonist[309]	Dectin-1 in NK cells	Improved long term survival of ovarian cancer patients
	Combination of trastuzumab and sialidases[310,311]	$HER2^+$ cancer cells	Desialylation of $HER2^+$ cancer cells and enhanced $CD8^+$ T cells and NK cell-mediated cytotoxicity
	Fluorinated analogues of sialic acid[312,313]	Sialylation in tumor cells	Precludes sialylation in cancer cells and causes the impairment of cell adhesion, migration, and tumor growth
	Treatment with 2-deoxyglucose (2DG)[314,316,317]	Tumor cells	Impairs *N*-glycan branching and improves CAR-T-mediated tumor killing and cytotoxicity
		T cells	Improves T cell anti-tumor response by increasing IL-2R retention and hampering Gal-3 binding
	Therapeutic antibodies (STM418)[222]	PD-1^+ T cells	Target glycans specifically on site N58 to increase anti-tumor responses and avoiding T-cell exhaustion
	Editing of N74 residue of PDCD1 gene[318]	PD-1^+ T cells	Removes PD-1 *N*-glycosylation and improves anti-tumor function of CAR-T cells
	Therapeutic antibodies (mbAb146)[319]	CTLA-4^+ T cells	Target CTLA-4 glycans to enhance T cell immune response

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levels of branched *N*-glycans in UC patients were associated with a positive therapeutic and disease outcome. Furthermore, it was shown that low levels of complex *N*-glycans combined with high C-reactive protein (CRP) levels, a marker of inflammation, dictates a poor prognosis in terms of non-response to therapy. This glyco-biomarker combined with CRP levels constitutes a powerful tool to improve prognosis and clinical-decision making process in IBD.[10] Similarly, abnormal mannosylated *N*-glycans on kidney biopsies of LN patients predict the development of chronic kidney diseases (CKD) with 93% specificity.[12] In context of cancer, certain molecular patterns associated with tumor *N*-glycome have been used as a tool for disease monitoring.[266–268]

Serum glycome has also been explored as important biomarkers of diseases. In fact, patients with active UC showed higher serological levels of L-selectin, Gal-1 and Gal-3 compared with healthy individuals.[269–271] Specifically, Gal-1 has discriminated UC patients from healthy individuals with 87% of specificity and similar sensitivity.[270] In SLE patients, a clinical trial is currently ongoing to evaluate the potential of Gal-9 as a biomarker of disease (NCT04558814).[173] In cancer, circulating Gal-1 is associated with higher recurrence and lower survival in melanoma patients treated with the anti-VEGF antibody bevacizumab.[272] Additionally, higher levels of α2,6-sialylated and branched structures identified in serum glycome of pancreatic cancer patients estimate a poorer prognosis.[266,267] CRC patients are enriched in multi-antennary and terminally sialylated or fucosylated, whose alterations follow the disease progression.[273] Moreover, the features of serum glycome could distinguish early non-small cell lung cancer patients from healthy individuals.[274]

Serum antibodies are also a fundamental piece in the pathogenesis of inflammatory and autoimmune diseases being an important tool to improve disease treatment and diagnosis. Firstly, antibodies against glycans or antibodies with altered glycosylation have been used in IBD diagnosis and to differentiate UC from CD patients. In fact, ASCA (anti-*Saccharomyces cerevisiae* antibodies), antibodies that target mannose-enriched glycans, are the most accurate biomarker of CD being widely associated with the severity of the disease.[275,276] Anti-glycoprotein 2 and anti-mannobioside carbohydrate IgGs[277,278] are also used for IBD diagnosis. Moreover, patients with CD exhibited increased agalactosylation IgGs levels, which positively correlates with levels of CRP, conferring a bad prognosis characterized by extensive and progressive disease.[141,279] Remarkably, a recent study identified a specific serum glycome signature on circulating IgGs detectable up to 6 years before CD diagnosis. The altered IgG Fc glycome signature (specifically agalactosylated IgG2 H3N4F1 glycoform), positively correlated with ASCA antibodies. This pathogenic Gycome-ASCA hub was demonstrated to be a promising new serum biomarker for CD prediction and a potential target for disease prevention.[280] Furthermore, in RA, glycan alterations of IgGs are present years before the onset of the disease.[281] Specifically, the ACPA-IgGs variable domain glycosylation (VDGs), enriched in bisecting and sialylated *N*-glycans structures, predicted the development of RA.[282] Interestingly, at the time of RA diagnosis, the lower abundance of ACPA VDGs

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was associated with drug-free remission.[283] In SLE and MS diseases, altered IgG *N*-glycome are also potential biomarkers of disease development.[284,285]

All these applications demonstrate how *N*-glycome assemble valuable information for the early detection and monitoring of inflammatory-, autoimmune-mediated diseases and malignancy.

3.6.2 Glycan-based therapies. Despite the recent advances in the therapeutic approaches for cancer, inflammatory and autoimmune diseases, considerable subset of patients remains refractory to conventional treatment or do not achieve sustained remission.[286–291] Remarkably, the possibility of reprogramming cellular glycosylation as a way of achieving homeostasis is highly appealing. In fact, GlcNAc supplementation has a positive effect on altered mucosal T cell response from UC patients, suggesting the potential immunomodulatory properties of GlcNAc as a therapeutic approach for patients with UC.[8,9] Moreover, *in vivo* supplementation with GlcNAc acted as an efficient prophylactic strategy by rescuing intestinal homeostasis through the enhancement of ILC3-IL-22 immunity and the levels of SCFAs.[292] In line, a pilot study using GlcNAc, has been developed in pediatric IBD.[293] Here, oral GlcNAc administration was reported to improve intestinal mucosal lining associated with clinical remission.[294] Moreover, *ex vivo* supplementation of GlcNAc in kidney biopsies recovered the homeostatic glycome, resulting in decreased infiltration of γδ T cells, a lower content of inflammatory cytokines, such as IL-17a, IFN-γ, TNF-α and up-regulation of IL-1012. Additionally, in MS and in IBD, *in vivo* GlcNAc administration was able to restore the branched *N*-glycans in TCR leading to a reduced T cell hyperactivation, avoiding inflammatory demyelination and disease severity.[8,9,44] In IIM patients, the *ex vivo* GlcNAc treatment of fresh muscle biopsies also attenuated the inflammatory response promoting a higher content of branched *N*-glycans and IL-4, and reducing the levels of IL-6.[14] In GNE myopathy patients, an open-label phase 2 study (NCT02346461) had showed that oral administration of ManNac was safe and effective in enhancing sialylation in these patients, where a clinical trial is currently active to test the effectiveness of this approach (NCT04231266).[162] All these evidence suggests glycosylation reprogramming through glycan supplementation as a promising therapeutic approach in chronic inflammation and autoimmunity, where two clinical trials are being assessed: a phase II/III trial (NCT01893606) and a phase III trial (NCT02504060).[293]

From an IgGs therapeutic perspective, *in vivo* administration of sialylated IgGs prevented the development of LN and arthritis by hampering the inflammatory responses mediated by pathogenic Th17 cells and autoantibodies.[295] Moreover, in mouse models of LN and RA, the administration of solubilized glycosyltransferases, namely B4GALT1 or ST6GAL1, aiming to recover the levels of galactose and sialic acid respectively, was able to convert the pathogenic IgG in an anti-inflammatory phenotype controlling the inflammatory response.[295] Immune complexes in SLE and joint swelling, as well as immune cell recruitment

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observed in RA were substantially improved through the treatment of autoantibodies with Endoglycosidase S, leading to the removal of *N*-glycans in IgGs.[296,297] Additionally, the depletion of autoreactive B cells was also considered a potential therapy mainly in autoimmunity, taking advantage of anti-CD20 monoclonal antibodies like Rituximab, ocrelizumab, and ofatumumab that are currently used for RA, MS and SLE patients.[298–302] Interestingly, a novel anti-CD20 monoclonal antibody with a distinct glycosylation profile of Fc-domain (G2 glycosylation pattern without fucose) was described as an optimized version of Rituximab, displaying an increased binding affinity for FcγRIIIα and a higher antibody-dependent cellular cytotoxicity in B cell lymphoma, being more effective in eliminating autoreactive B cells.[303,304] In SLE, treatment with neuraminidase has restored the Gal-1 binding to T cells, promoting an anti-inflammatory environment.[172] In a mice model of RA, the administration of Gal-1 resulted in clinical suppression due to anti-inflammatory responses mediated by Th2.[305] On the other hand, the depletion of Gal-3 resulted in fewer clinical manifestations and lower levels of inflammation in RA[178] and MS[166] mice models.

The therapeutic potential of galectins and siglecs in TME has also been studied. In fact, targeting Gal-1 binding improved $CD8^+$ T cell infiltration and its cytotoxic capacity, enhancing tumor killing.[306–308] Similarly, the blocking of Gal-3 with galactomannan, GM-CT-01, in patient-derived TILs was shown to be effective in improving $CD8^+$ T cell anti-tumor response.[309] Additionally, Gal-3 inhibitor, GR-MD-02, was also tested in combination with immune blockade antibodies in solid tumors clinical trials.[310] Regarding siglecs, monoclonal antibodies were tested against Siglec-7, -9 and -15.[196,200,311–313] Moreover, CAR-T cells engineered to express Siglec-7 and -9 demonstrated enhanced anti-tumor capacity, through improved recognition of sialoglycans on tumors.[314] In terms of GBP interactions, the most widely approach considered is the targeting, either with antibodies or binding peptides, of tumor antigens to GBP on innate immune cells to generate an enhanced immune response.[315–318] The use of GBP agonists has also been explored as an anti-tumor therapy. For instance, the use of an agonist of Dectin-1, β-Glucan, showed promising results in combination with chemotherapy, improving long term survival of OC patients.[319]

Given the prominent role of glycans in the treatment of chronic inflammation and autoimmune diseases, also, in cancer, aberrant *N*-glycosylation poses a promising target in cancer treatment and precision medicine. One notable case is the conjugation of the anti-HER2 monoclonal antibody trastuzumab with a sialidase, which specifically targets HER2+ cells for desialylation. This glycoreprogramming will favor $CD8^+$ T cells and NK cell-mediated cytotoxicity by precluding Siglec engagement.[320,321] Furthermore, fluorinated analogues of sialic acid, which inhibit sialylation, abrogated *in vitro* migration and *in vivo* tumor growth.[322,323] *N*-glycan branching is another relevant feature of tumors with therapeutic potential. Indeed, treatment with 2-deoxyglucose (2DG), a glucose analogue that is converted in D-mannose entering *N*-glycosylation pathway, sensitize pancreatic tumor cells to CAR-T cell killing

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in vitro and *in vivo*.[324] Thus, glycan reprogramming may assist in treating solid tumors that are unresponsive to therapy such as CAR-T cell therapy. Other aberrant patterns of *N*-glycosylation, such as enhanced fucosylation have also been explored with the same goal.[325] Moreover, and as mentioned previously, T cells are highly regulated by *N*- glycosylation, which makes them a potential target to reinvigorate T cell function against tumors. To alter the *N*-glycosylation profile of T cells, Sasawatari and colleagues treated human T cells with 2DG showing an improved T cell anti-tumor immune response, characterized by increased cytotoxicity, IL-2R surface retention and inhibition of Gal-3 binding.[326] In the same line, Toyofuku and colleagues treated CAR-T cells with 2DG and observed that these cells had less binding affinity to immune-inhibitory receptors, presenting an increased cytotoxicity.[327] In particular, *N*-glycosylation of T cell receptors can also be target, as for example *N*-glycosylation of PD-1 and CTLA-4. For instance, therapeutic antibodies (STM418) to target the N58-glycan presented on PD-1 were developed and showed an enhanced induction of anti-tumor responses.[230] Furthermore, by editing the coding sequence of the N74 residue of *PDCD1* gene to remove PD-1 *N*-glycosylation reduced its expression on cell surface, improving anti-tumor activity of CAR-T cells.[328] Likewise, therapeutic antibodies against the *N*-glycosylated forms of CTLA-4 (mbAb146) were also developed to increase T cell efficacy against tumors.[329]

Overall, and building upon the master regulatory properties of *N*-glycans in the immunopathogenesis of major diseases, their role as promising diagnostic and prognostic biomarkers, as well as potential therapeutic targets hold the promise of revolutionizing the glycomedicine field in a near future.

4 Conclusion

The interaction between glycans and GBPs constitutes a fundamental hub that tightly modulates host immune cell response both in homeostasis and in disease context. In this chapter, we comprehensively discussed how glycans and GBPs integrate the canonical circuits that regulate both innate and adaptive immune responses. We discussed the main *N*-glycosylation alterations that occur in the transition from health to disease contexts, specifically in intestinal inflammation, autoimmunity, and malignant transformation. Additionally, we addressed the potential clinical applications of *N*-glycans and GBPs as promising clinical glyco-biomarkers and novel targets for the development of new glycan-based therapies for improving personalized medicine, as described in Tables 1 and 2.

Acknowledgements

This work was co-funded by the European Union (ERC, GlycanSwitch, Grant Agreement N° 101071386) and by the (GlycanTrigger, Grant Agreement N°: 101093997). Views and opinions expressed are however those of the author(s) only and do not necessarily reflect those of the

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European Union or the European Research Council Executive Agency. Neither the European Union nor the granting authority can be held responsible for them. SSP acknowledges funding by the US Department of Defense, US Army Medical Research Acquisition Activity, FY18 Peer Reviewed Medical Research Program Investigator-Initiated Research Award (award number W81XWH1920053). SSP also acknowledges funding from European Crohn's and Colitis Organisation (ECCO) Pioneer Award 2021; the International Organization for the study of Inflammatory Bowel Disease (IOIBD) and the Portuguese Foundation for Science and Technology – FCT (EXPL/MED-ONC/0496/2021). SSP acknowledges 2022 Lupus Research Alliance (LRA Lupus Innovation Award) and Liga Portuguesa Contra o Cancro LPCC and MULTICARE for funding (C 3035006C17). JG acknowledges European Society of Clinical Microbiology and Infectious Diseases (ESCMID Research Grant 2022) and European Crohn's and Colitis Organization for funding (ECCO Grant 2023). IA acknowledges funding from Ordem dos Médicos and BIAL Foundation (Maria de Sousa Award 2023), from Group of Studies for Autoimmune Diseases and from European Society for Clinical Investigation (ESCI Exploratory Research Grant 2024).AF acknowledges Portuguese Foundation for Science and Technology (FCT) for funding (2022.01422.PTDC). CMA [BD/07357/2021]; BSP [UI/BD/151550/2021]; ELG [UI/BD/152866/2022]; CSR [2020.08422.BD]; JG [2020.00088.CEECIND] and IA [2022.00337.CEECIND] received funding from the FCT of the Portuguese Ministry of Science, Technology and Higher Education.

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